

HS Harz FuSa, Friedrichstraße 57

Airbag ECU

Messung und Verarbeitung der Sensoreingänge zur
Auslösung des Airbags
Measurement and Processing of Sensor Signals to Process
an Airbag Operation

Phase 3: Functional Safety Concept (FSC)

Version: 1.00

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1 Meetings within the project

Topic	Time	Room	Participants
Item Definition	16:00	Zoom	ALL
Bericht:	<u>none</u>		

2 Unresolved topics

No unresolved topics

3 Glossar

Tabelle 1: Glossar

Begriff / Term	Erklärung / Explanation
ACU	Airbag Control Unit.
ADC	Analog to digital converter value read from an analog sensor input.
ASIL	Automotive Safety Integrity Level used by ISO 26262 for risk classification.
CAN	Controller Area Network, a vehicle communication bus.
ECU	Electronic Control Unit.
EUC	Equipment under Control.
FuSa	Functional Safety.
HW	Hardware.
QM	Quality Management level below ASIL.
SIL	Safety Integrity Level.
SW	Software.
ACU	Airbag Control Unit.
ADC	Analog-to-digital converter value read from an analog sensor input.
ASIL	Automotive Safety Integrity Level used by ISO 26262 for risk classification.
CAN	Controller Area Network, a vehicle communication bus.
ECU	Electronic Control Unit.
EUC	Equipment under Control.
FuSa	Functional Safety.
HW	Hardware.

4 Functional Safety Concept (FSC)

Safety Goals represent top-level requirements derived as the inversion of identified hazards. These goals establish the mandate for the system's Safe State which for the Airbag ECU is typically deployment inhibited or firing loop isolation.

4.1 Safety Goals (SG1 to SG4)

Tabelle 2 Safety Goals

SG ID	Safety Goal	ASIL	Safe State
SG1	Prevent unintended deployment.	ASIL D	Deployment inhibited / firing loop isolation.
SG2	Ensure deployment occurs within the FTTI during a valid crash.	ASIL D	Deployment inhibited / firing loop isolation.
SG3	Suppress passenger airbag deployment when a baby seat is detected.	ASIL C	Deployment inhibited / firing loop isolation.
SG4	Ensure correct airbag selection based on impact zone.	ASIL D	Deployment inhibited / firing loop isolation.

SG3 is a vital logic requirement: the safety concept must ensure that a Suppress command overrules Fire logic when specific occupancy criteria are met. Any failure in this suppression logic is a direct violation of the Functional Safety Concept.

4.2 Technical Safety Mechanisms and Diagnostics

The Technical Safety Concept (TSC) translates Functional Safety Requirements into hardware diagnostics. To satisfy the ASIL D mandate, the following mechanisms are implemented to protect the SPI and processing paths.

Tabelle 3 Technical Safety Mechanisms and Diagnostics

Safety Mechanism	Description
CRC Checks	Applied to the transmit/receive shift registers to detect data corruption in the SPI path.
Interrupt Count and Timeliness Checks	Detects missing/spurious interrupts or protocol delays, protecting against failures in the state machine or interrupt generation logic.
Register Toggle Verification	Validates that configuration settings (CPOL/CPHA) and mode selections have not been corrupted by random hardware failures.

4.3 Validation via Fault Injection (FMEDA)

As per ISO 26262-9, the effectiveness of these mechanisms is validated using FMEDA-based fault injection. Two simulation flows are used:

Tabelle 4 Validation via Fault Injection (FMEDA)

Simulation Flow	Description
Concurrent Run	A high-throughput simulation identifying Dangerous Detected (DD) faults across large groups of faults.
Serial Run	A high-precision simulation targeting Dangerous Undetected (DU) faults to ensure they are either detected by checkers or reclassified as Safe after detailed netlist analysis.

4.4 Hardware Metrics Achievement

Based on the IJERT FMEDA results, a Diagnostic Coverage (DC) of 97.2% was achieved for single-point faults in the SPI block. This high coverage for the communication peripheral is the technical enabler that allows the entire ECU to meet the 99% Single Point Fault Metric (SPFM) required for ASIL D compliance. This rigorous validation confirms that the Airbag ECU adheres to the absolute mandates of ISO 26262 Parts 5 and 9.

Tabelle 5 Hardware Metrics Achievement

Metric	Value Achieved	Requirement
Diagnostic Coverage (DC) — SPI single-point faults	97.2%	ISO 26262-5
Single Point Fault Metric (SPFM)	99%	ASIL D (mandatory)

5 Figure and table listing

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6 List of dependent documents

No.	Dokument name	Version	Date	File name
1	Hazard Analysis and ASIL Determination for Airbag ECU	V02	02.06.2026	Hazard Analysis and ASIL Determination for Airbag ECU V02.pdf
2	Hazard Analysis and ASIL Determination for Airbag ECU	V01	07.05.2026	Hazard Analysis and ASIL Determination for Airbag ECU_V01.pdf
3	Airbag ECU ASIL Code Division	V02	02.06.2026	Airbag_ECU_ASIL_Code_Division_V02.docx
4	Airbag ECU Hardware Concept Design Report	V02	27.05.2026	AirbagECU_HW_Concept_Design_Report_V02.pdf

7 Literature

- [1] M. Skutek, M. Mekhaieel, and G. Wanielik, A precrash system based on radar for automotive applications, in IEEE IV2003 Intelligent Vehicles Symposium, Columbus, OH, USA, 2003, pp. 37-41. doi: 10.1109/IVS.2003.1212879.
- [2] H. Komaki, A. Kitabatake, T. Koyama, T. Tabata, and H. Konishi, Denso-Ten Technical Journal. Available: <https://www.denso-ten.com/business/technicaljournal/pdf/18-4.pdf>
- [3] RDAIRBAGPSI5 Airbag Reference Platform, Freescale Semiconductor, Inc. Available: <https://community.nxp.com>
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- [5] embien, Guide to Airbag Control Unit ACU - Enhancing Automotive Safety. Available: <https://www.embien.com/automotive-insights/guide-to-airbag-control-unit-acu-enhancing-automotive-safety>
- [6] element14 Community, Freescale automotive DSI airbag system, Jun. 20, 2012. Available: <https://community.element14.com>
- [7] audioholic, Technically understanding Airbag systems and SRS, Team-BHP.com, Aug. 12, 2017.

8 Version history

Document name	date	Change log
Airbag_ECU_Phase_3	02.06.2026	v1.00 Initial version. Safety Goals from HARA V02 (02.06.2026). Technical Safety Mechanisms and DC 97.2% from HARA V01 (07.05.2026). ASIL Code Division V2 (02.06.2026) referenced for FSC.